PROGRAMMING WITH PYTHON I

Master in Computer Science & Business Technology MCSBT SEP-2026 S-1

Area Computer Science
Number of sessions: 20
Term: Term 1
Category: regular
Language: English

Professor: **GUSTAVO MARTIN VELA**

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Gustavo Martín Vela is a Telecomm Engineer and Fine Arts Master with a passion for scientific computing, programming, hardware devices, education, and outreach. Currently, he works as Data Engineer Lead at Technology dept. in INDITEX, a company on a mission to give to the customers the best dressing experience based on a data driven model. He has worked as data engineer, data scientist, data architect and researcher in the insurance, retail, oil & gas and telecom industries since 2010, and as a Big Data and Data Science trainer for several private entities.

CORPORATE EXPERIENCE

- Cloud Data and AI Engineer Lead, INDITEX, Spain, 2025 – Present
- Cloud Data and AI Engineer Lead, LOVES, USA, 2023 – 2025
- Cloud Data Engineer Lead, INDITEX, Spain, 2021 – 2023
- Data Engineer / Data Engineer Lead, MASMOVIL Group, Spain, 2019 – 2021
- Data Scientist / Data Engineer Lead, IKEA, Spain – Sweden, 2017 – 2019
- Data Engineer / Data Scientist, TELEFONICA, Spain, 2014 – 2017
- Business Analyst Engineer, TELEFONICA, Spain, 2013 – 2014
- Data Researcher, TELEFONICA RESEARCH, Spain, 2011 – 2013
- Research Assistant, Deustotech, Spain, 2010 – 2011

ACADEMIC EXPERIENCE

- Adjunct Professor of technology, IE University, Spain, 2022 – Present\
- Data Science Instructor Lead, The Bridge, Spain, 2021 – Present
- Big Data Instructor, Private Entities, Spain, 2016 – Present

Office Hours

Office hours will be on request. Please contact at:

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SUBJECT DESCRIPTION

This course is your launchpad into the world of programming, using Python, one of the most versatile and most used languages in the world. You'll learn by doing. We' ll cover the essential concepts like variables, loops, functions, and data types through practical examples you'll actually use, from automating everyday tasks to analyzing simple datasets.

LEARNING OBJECTIVES

- Collaborate effectively with AI tools and human peers to solve problems and generate innovative solutions.

- Build small, functional Python programs from scratch
- Understand how code solves problems, a skill that transfers to any programming language
- Read and write clean, reusable code using best practicesTackle future technical courses with confidence
- Productivize simple python code in an automatic way

TEACHING METHODOLOGY

IE University teaching method is defined by its collaborative, active, and applied nature. Students actively participate in the whole process to build their knowledge and sharpen their skills. Professor's main role is to lead and guide students to achieve the learning objectives of the course. This is done by engaging in a diverse range of teaching techniques and different types of learning activities.

Learning Activity	Weighting
Lectures	20.0 %
Exercises in class, Asynchronous sessions, Field Work	20.0 %
Group work	30.0 %
Individual studying	30.0 %
TOTAL	100.0 %

AI POLICY

Critical GenAI use is encouraged
In this course, the use of generative artificial intelligence (GenAI) is encouraged, with the goal of developing an informed critical perspective on potential uses and generated outputs. However, be aware of the limits of GenAI in its current state of development:

- If you provide minimum effort prompts, you will get low quality results. You will need to refine your prompts to get good outcomes. This will take work.
- Don't take ChatGPT's or any GenAI's output at face value. Assume it is wrong unless you either know the answer or can cross-check it with another source. You are responsible for any errors or omissions. You will be able to validate the outputs of GenAI for topics you understand.
- AI is a tool, but one that you need to acknowledge using. Failure to do so is in violation of academic honesty policies. Acknowledging the use of AI will not impact your grade.

Suggested format to acknowledge the use of generative AI tools:
I acknowledge the use of [AI systems link] to [specify how you used generative AI]. The prompts used include [list of prompts]. The output of these prompts was used to [explain how you used the outputs in your work].

If you have chosen not to include any AI generated content in your assignment, the following disclosure is recommended:
No content generated by AI technologies has been used in this assignment.

PROGRAM

SESSION 1 (LIVE IN-PERSON)

Professional Python Foundations and Functional Programming

Set up a modern, production-ready Python project with venv or uv, version control, and a logical folder structure. Then jump straight into writing clean, reusable functions with type hints and docstrings.

We'll cover:

- Create a src/ based project folder with virtual environment using venv or uv
- Initialize Git, .gitignore, and README.md
- Write a multifunction calculator with type hints

SESSION 2 (LIVE IN-PERSON)

Data Types, Control Flow, and Error Handling

Deepen your grasp of core Python data types, conversions, and error handling. Practice writing defensive code with nested conditionals and loops.

You'll build small programs like:

- Build a menu-driven console application
- Create a quiz with scoring and retries
- Write a unit converter with exception handling
- Refactor repetitive logic into clean functions

SESSION 3 (LIVE IN-PERSON)

Advanced Data Structures

Explore powerful data organization with lists, dictionaries, tuples, sets, and nested structures to build more complex and realistic applications.

Expect to create things like:

- Build a contact book with nested dictionaries
- Parse a JSON like dict structure
- Develop a student grade tracker

SESSION 4 (LIVE IN-PERSON)

Python Idioms and Patterns

Learn the most common and efficient "Pythonic" idioms and patterns to write clean, elegant, high-performance code.

We'll apply this to:

- Rewrite loops using list comprehensions
- Combine two lists with zip
- Use lambdas with map and filter

SESSION 5 (LIVE IN-PERSON)

File I/O and Structured Data

Persist data using text files, JSON, and CSV, and learn how to design file-based mini databases.

You'll practice:

- Build a to-do list app storing tasks in JSON
- Import a CSV of sample expenses
- Write a CSV export function

- Validate file content before loading

SESSION 6 (LIVE IN-PERSON)

Advanced String Handling and Data Cleaning

Go beyond basic string methods to master advanced text manipulation, pattern matching, and data cleaning techniques that are essential in real world applications. Learn to sanitize user input, handle encoding issues, and parse messy data sources.

You'll build programs like:

- Clean up a messy CSV file with irregular spacing and case
- Validate email addresses with regex
- Parse phone numbers to a consistent format
- Normalize text input from users (capitalization, whitespace)
- Write a reusable data cleaning module for later projects

SESSION 7 (LIVE IN-PERSON)

Command Line Tools and Automation

Use Python to automate tasks and interact through the command line, including parsing arguments and running scripts from the terminal.

Real-world uses include:

- Automate budget data backups with a script
- Parse CLI arguments with argparse
- List expenses from the command lineSchedule a script to run with cron or Task Scheduler

SESSIONS 8 - 9 (LIVE IN-PERSON)

Advanced Object-Oriented Programming (OOP) Techniques

Transition from procedural code to an object-oriented mindset using classes, objects, and encapsulation to model real-world problems. Apply inheritance, class methods, static methods, and polymorphism to make flexible, powerful class structures.

You'll learn how to:

- Extend Transaction with a recurring transaction subclass
- Add a class method to summarize expenses

- Use static methods for validation
- Refactor your contact book to OOP style

SESSION 10 (LIVE IN-PERSON)

Lab: Building an Interactive Personal Budget Manager

Goal: Combine fundamentals like variables, conditionals, loops, functions, file handling, into one project creating a program that lets users:

- Input income and multiple expense entries
- Categorize expenses (e.g., Food, Transport, Entertainment)
- Save and load data from a CSV or JSON file
- Use conditionals to validate inputs (e.g., no negative amounts)
- Implement loops to allow multiple entries without restarting the program
- Create functions to organize the code (e.g., for input, calculation, file I/O)
- Provide summary reports like total expenses, balance remaining, and category-wise breakdown

SESSION 11 (LIVE IN-PERSON)

Midterm practical exam

- Open book, students are allowed to use slides from class and all code from class, but no other resources.
- Multiple choice/multiple answer. Conducted through Blackboard.

SESSION 12 (LIVE IN-PERSON)

Working with APIs and JSON Data

Learn to connect to the internet through REST APIs, process JSON responses, and handle authentication.

We'll cover:

- Fetch live currency exchange rates from an API
- Parse and store JSON responses
- Handle HTTP errors and timeouts
- Integrate live rates into your budget app

SESSIONS 13 - 14 (LIVE IN-PERSON)

Organizing Code with Modules and Leveraging Popular Python Libraries

When your programs grow larger, keeping code organized and using powerful tools is key. This session teaches you how to modularize your code with modules and introduces some of the most widely used Python libraries that will help you solve real-world problems efficiently.

We'll cover:

- Creating and importing your own modules for cleaner, reusable code
- How to install and use external libraries with pip

Popular libraries you'll get familiar with:

- requests — for accessing and interacting with web APIs
- pandas — for powerful data manipulation and analysis
- duckdb — for running SQL queries on your data right inside Python
- matplotlib & seaborn — for creating data visualizations
- flask — for building simple web applications and APIs
- fastapi — for creating powerful APIs to serva data over the web
- beautifulsoup4 — for scraping and parsing HTML web pages
- streamlit — for publishing and interacting with results and chart in the web

SESSION 15 (LIVE IN-PERSON)

Web Development with Flask

Learn to build a web app using Flask to serve HTML pages and handle forms, the foundation for modern Python web dev.

Practical exercises:

- Build a simple web route showing expense history
- Create a form to add new transactions
- Deploy a basic app locally

SESSION 16 (LIVE IN-PERSON)

APIs with FastAPI

Go beyond Flask with modern, high-performance web APIs using FastAPI, supporting JSON data and modern features.

Practical exercises:

- Create an API endpoint to serve budget data
- Add routes to post new transactions
- Document your API with built-in OpenAPI

SESSION 17 (LIVE IN-PERSON)

Data Analysis with pandas and duckdb

Leverage pandas and duckdb to analyze, transform, and summarize data programmatically, with best practices for real data pipelines.

Practical projects:

- Load a CSV of expenses into pandas and duckdb and comparing both approachs
- Filter and group expenses by category
- Aggregate monthly totals
- Merge two datasets

SESSION 18 (LIVE IN-PERSON)

Interactive Dashboards with Streamlit

Bring your analysis to life with interactive dashboards and simple user interfaces in Streamlit.

Practical projects:

- Build an expense dashboard with filters
- Show charts in Streamlit
- Add live updates to your budget data
- Package a Streamlit app for local deployment

SESSION 19 (LIVE IN-PERSON)

Advanced Lab: Data Explorer and Visualizer with API Integration

Goal: Combine all skills in one final, portfolio-grade project: a personal finance manager with a web UI, API endpoints, data analysis, and interactive charts.

Lab Tasks:

- Public API to fetch live data with requests
- Store the retrieved data in JSON or CSV files for reuse
- Load and analyze data using duckdb (e.g., filtering, aggregation)
- Create visualizations with streamlit (charts, graphs)
- Package functionality into functions or modules for readability and reuse

Learnings:

- Build the end-to-end app

- Integrate data storage, visualization, and APIs
- Package the project for GitHub

SESSION 20 (LIVE IN-PERSON)

Final exam
Multiple choice/multiple answer and writing questions related to code. NO conducted through Blackboard, conducted in class.

EVALUATION CRITERIA

This course uses a balanced assessment structure designed to evaluate both individual achievement and collaborative learning. Students will be assessed through a **final exam (30%)**, **individual work (20%)**, **group work (30%)**, **class participation (10%)**, and **intermediate tests (10%)**. Together, these components measure students' understanding, practical application, teamwork, and active engagement throughout the course. A central learning objective is to develop the ability to collaborate effectively with AI tools and human peers to solve problems and create innovative solutions.

criteria	percentage	Learning Objectives	Comments
Final Exam	30 %	- Collaborate effectively with AI tools and human peers to solve problems and generate innovative solutions.	
Individual work	20 %	- Collaborate effectively with AI tools and human peers to solve problems and generate innovative solutions.	
Group Work	30 %	- Collaborate effectively with AI tools and human peers to solve problems and generate innovative solutions.	
Class Participation	10 %	- Collaborate effectively with AI tools and human peers to solve problems and generate innovative solutions.	
Intermediate tests	10 %	- Collaborate effectively with AI tools and human peers to solve problems and generate innovative solutions.	

FAILING GRADE AND REASSESSMENT

When students receive a Fail in a course, they have the opportunity to present themselves for reassessment in order to earn the necessary credits toward graduation.

The reassessment of students should be scheduled between 5 and 10 working days after the review session takes place.

Grades for the reassessment are limited to a Low Pass and Fail.

Both, the initial Fail as well as the grade of the reassessment remain on the transcript. For the purpose of calculating the GPA however, only the grade of the reassessment is to be considered. Students receiving a failing grade in the reassessment of a course will not be able to continue in the program.

BIBLIOGRAPHY

Recommended

- Al Sweigart. (0). *Automate the Boring Stuff with Python.* ISBN 1593279922 (Digital)

- Paul Barry. (0). *Head First Python: A Brain-Friendly Guide.* O'Reilly Media. ISBN 1492051292 (Digital)

- Allen B. Downey. (2015). *Think Python: How to Think Like a Computer Scientist.* 2nd. O'Reilly Media. ISBN 9781491939369 (Digital)

SOFTWARE

- *ChatGPT API TOKEN.* OpenAI. N/C.

BEHAVIOR RULES

Please, check the University's Academic Integrity Policy here. The Program Director may provide further indications.

ATTENDANCE POLICY

Please, check the University's Attendance Policy here. The Program Director may provide further indications.

ETHICAL POLICY

Please, check the University's Student Code of conduct here. The Program Director may provide further indications.